Wireless Transmission
1.1 How can the capacity be increased in interference-limited systems when the allocated network bandwidth is fixed? (1 point)

1.2 State an advantage of spreading a narrow-band signal. (1 point)

1.3 Impact of the SINR on the bit error rate for the case of 160AM modulation: depict examples for the impact of the channel quality on the phase diagrams. (3 points)

1.4 What-is the beneft of dynamic channel.adaptation? For which scenarios is it fast channel adaptation especially beneficial? (2 points)

2 Medium Access Schemes
2.1 State the advantages of reservation-based MAC schemes with respect to ...

- Occurrence of collisions (1 point)
- Utilization of the radio resource as whole (1 point)
- Access latency in heavily loaded system (1 point)

2.2 What is a unique challenge in CDMA (Code Division Multiple Access) that is not present in TDMA or FDMA? (1 point)

2.3 mention advantages of Centralised MAC schemes, for instance, as employed with cellular networks. (2 points)

2.4 Solution of the distributed access coordination problem: Describe how the RTS/CTS scheme tries to solve the problem. (3 points)

3 IEEE 802.11

3.1 How can NAV be used to avoid collisions in the Distributed Coordination Function (DCF)?

3,2 What is the central idea for incorporating energy saving in IEEE 802.11? How is it implemented in Ad-hoc networks? (2 points)

3.3 Which among the two schemes, PCF or DCF, can ensure better QoS for offered service Provide an explanation for your answer. (2 points)

4 Mobility Management
4.1 Layer 2 mobility: how is mobility handled if a node moves within an IP subnet
4.2 What are the drawbacks of implementing the triangular routing scheme in Mobility Management

4.3 Mobile IP:
Reverse tunnelling diagram question
4.4 Which protocol can be used to acquire a CoA Tor Mobile IP (1point)

5 Transport Protocols
5.1 What is the purpose of TCP ports? (1 point)
5.2 How TCP protocol ensures reliability? (2 points)
5.3 In the case of wireless communications, can TCP incorrectly detect congestion? Provide an explanation for your answer. (2 points)

5.4 One of the TCP/IP limitations is Head of Line (Hol) blocking. What is it and when does it happen? (2 points)

6 QoS
6. 1 List the four QoS attributes in communication systems. (2 points)

6.2 What is the functional difference between RED and WRED? (1 point)

6.3 What is the purpose of the token bucket scheme? (1 point)
6.4 The IEEE 802.11e standard provides an extension of the DCF scheme to support QoS. Which of the three Reneral strategies for QoS provisioning is applied with this extension? (1 point)

6.5 Is there a trade-off between reliability and latency? Justify your answer. (2 points)

7 Self-Organization
7.1 Name the main properties in which self-organized networks differ from centralized networks. (2 points)

7.2 Name two protocols with their related mechanisms that apply principles of self organization. (2 points)

7.3 Discuss operation at the Edge of Chaos. (2 points)

7.4 Why is it particularly difficult to develop and test self-organized systems? (1 point)

8 Cellular Networks
8.1 For what type of IP traffic can the TDD duplex scheme utilize the spectrum for UL and DL more efficiently? Why? (2 points)

8.2 What is the benefit of soft handover, and how does it relate to the concept of macro diversity? (2 points)

8.3 Consider a given frequency range:
• a. What is the purpose of spatial reuse in cellular networks? (1 point)
b. How is the frequency reuse factor, e.g., 3 vs. 7, related to overall system capacity and interference? (1 point)

9 Ad-hoc Networks

9.1 In the case of a low mobility and less dynamic ad-hoc network scenario with sporadic traffic only, which class of routing approach is more suitable? Provide the pros and possible cons of your proposed approach? (2 points)

9.2 AODV:
when the RREG message initiated not just forwarded

9.3 What is the fundamental idea behind geographica rouling afortms based on simple greedy approaches? (2 points)

10 Cognitive Radio Networks

10.1 Mention an advantage and a disadvantage of Cooperative Detection strategies for spectrum sensing in Cognitive Radio Networks? (2 points)

10.2 Explain the principle of the underlay mode of operation for spectrum sharing, (2 point)
10.3 Describe a challenge of spectrum sharing in overlay mode. You may use an example (2 points)
10.4 What is the idea of using software-defined radios (SDRs)? (1 point)

10.5 What are the reasons for the limited practical application of SDRs in industry? (2 points)